Genome-wide fusion-hotspot cohort scan: msk_impact_50k_2026

544 of 3919 genes passed the >= 5-patient recurrence gate; 4 FDR-significant (q<0.05).

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Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
NTRK3	auto	ok	71	94.83	94.83	7.663e-09	yes	0.3943
ROS1	auto	ok	204	59.02	87.7	0.0002077	yes	0.3874
ETV6	auto	ok	213	71.11	75.56	0.002189	yes	0.4489
FLI1	auto	ok	121	61.86	5.932	0.007702	yes	0.5808
RET	curated	ok	219	75.26	92.27	0.09074	no	0.4345
FGFR2	auto	ok	188	79.41	84.56	0.09074	no	0.3373
ERG	auto	ok	863	30.2	96.07	0.2116	no	0.4484
EWSR1	auto	ok	415	65.33	11.17	0.2116	no	0.2932
ALK	curated	ok	321	83.09	95.96	0.2116	no	0.4963
TP53	auto	ok	258	8.511	25.53	0.2116	no	0.2284
BRAF	curated	ok	247	84.36	91.06	0.2116	no	0.3299
EML4	auto	ok	224	87.89	53.36	0.2116	no	0.6667
FGFR3	auto	ok	194	48.68	93.42	0.2116	no	0.4924
TACC3	auto	ok	138	50	48.48	0.2116	no	0.5678
NOTCH1	auto	ok	115	23.4	59.57	0.2116	no	0.3398
NTRK1	curated	ok	113	41.03	82.05	0.2116	no	0.2923
NAB2	auto	ok	111	34.72	68.06	0.2116	no	0.3891
CDK12	auto	ok	99	16.28	37.21	0.2116	no	0.2757
TERT	auto	ok	92	0	68.42	0.2116	no	0.3993
MET	auto	ok	73	51.52	72.73	0.2116	no	0.285
POLE	auto	ok	73	18.18	31.82	0.2116	no	0.2045
STAT6	auto	ok	60	43.4	5.66	0.2116	no	0.3594
DNAJB1	auto	ok	54	64.1	2.564	0.2116	no	0.5432
RNF43	auto	ok	48	28.57	33.33	0.2116	no	0.2448
KIAA1549	auto	ok	44	90.91	50	0.2116	no	0.5302
POLD1	auto	ok	36	12.5	37.5	0.2116	no	0.2327
RBM10	auto	ok	36	42.86	71.43	0.2116	no	0.347
PRKACA	auto	ok	35	68.57	82.86	0.2116	no	0.6812
ATF1	auto	ok	30	70	90	0.2116	no	0.3942
BICC1	auto	ok	29	96.43	67.86	0.2116	no	0.4292

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
RPS6KA4	auto	ok	21	38.46	53.85	0.2116	no	0.2137
LMNA	auto	ok	17	46.15	38.46	0.2116	no	0.4387
AGK	auto	ok	14	92.86	14.29	0.2116	no	0.622
TPM3	auto	ok	14	78.57	64.29	0.2116	no	0.5362
NR4A3	auto	ok	7	57.14	71.43	0.2116	no	0.3723
EMID1	auto	ok	5	75	25	0.2116	no	0.4674
EGFR	auto	ok	466	50.91	74.55	0.2368	no	0.3219
CREB1	auto	ok	7	57.14	71.43	0.2368	no	0.3814
EZH1	auto	ok	30	0	18.18	0.2529	no	0.3722
TSC2	auto	ok	88	17.65	23.53	0.2717	no	0.2546
PRKD1	auto	ok	16	50	25	0.2717	no	0.3033
ASPSCR1	auto	ok	20	70	20	0.2825	no	0.4852
SEC16A	auto	ok	17	23.53	41.18	0.2825	no	0.3708
AGO2	auto	ok	49	26.09	30.43	0.3155	no	0.326
NCOR1	auto	ok	80	0	31.58	0.3284	no	0.3513
KDR	auto	ok	28	11.11	33.33	0.3284	no	0.2071
SLX4	auto	ok	28	26.32	57.89	0.3284	no	0.383
CASP8	auto	ok	19	22.22	55.56	0.3284	no	0.2601
CDKN2B	auto	ok	40	16.67	16.67	0.3606	no	0.3909
CD74	auto	ok	74	58.21	88.06	0.3682	no	0.432
NOTCH2	auto	ok	92	3.571	50	0.3744	no	0.2303
CCDC6	auto	ok	53	92.45	1.887	0.3744	no	0.4276
DNMT3A	auto	ok	32	36.36	36.36	0.3744	no	0.3121
SUFU	auto	ok	17	25	25	0.3964	no	0.2274
RPTOR	auto	ok	43	23.08	15.38	0.4183	no	0.2673
BCAS3	auto	ok	18	6.25	43.75	0.4183	no	0.2759
PAX8	auto	ok	43	30	80	0.4605	no	0.2854
INPPL1	auto	ok	39	28.57	50	0.4733	no	0.2332
ACOXL	auto	ok	8	12.5	50	0.4936	no	0.3113
KIF5B	auto	ok	95	88.54	96.88	0.5017	no	0.6266
TOP1	auto	ok	31	0	50	0.5304	no	0.4077
FGFR1	auto	ok	64	30.43	47.83	0.5489	no	0.2254
MAP3K1	auto	ok	42	10	30	0.5489	no	0.2556
CARM1	auto	ok	39	22.22	27.78	0.5489	no	0.2168
PRKAR1A	auto	ok	12	33.33	16.67	0.5489	no	0.2584
CREBBP	auto	ok	121	13.04	32.61	0.55	no	0.2359
ERBB3	auto	ok	45	11.11	55.56	0.5808	no	0.219

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
STK11	auto	ok	97	4.348	30.43	0.5929	no	0.2657
CSDE1	auto	ok	31	0	60	0.6014	no	0.4667
TSC1	auto	ok	28	0	57.14	0.6014	no	0.4304
NRG1	auto	ok	16	53.33	86.67	0.6083	no	0.2663
FBXL20	auto	ok	8	28.57	28.57	0.6083	no	0.3592
FLT4	auto	ok	35	14.29	42.86	0.6245	no	0.2201
KMT2C	auto	ok	134	6.25	46.88	0.673	no	0.2041
FH	auto	ok	29	6.25	12.5	0.673	no	0.2761
ABL1	auto	ok	21	0	22.22	0.673	no	0.3956
DGKH	auto	ok	7	0	28.57	0.673	no	0.4272
ELF3	auto	ok	47	27.27	45.45	0.6751	no	0.2607
RAD51B	auto	ok	16	50	50	0.6751	no	0.3065
ARID1B	auto	ok	72	0	53.85	0.6807	no	0.4781
TFE3	auto	ok	45	56.76	78.38	0.6807	no	0.3513
AKT2	auto	ok	22	10	20	0.6807	no	0.2038
TTC28	auto	ok	15	16.67	58.33	0.6807	no	0.2223
GOPC	auto	ok	6	75	50	0.6807	no	0.4432
SND1	auto	ok	17	94.12	17.65	0.7018	no	0.3638
FAT1	auto	ok	95	0	27.27	0.7034	no	0.4095
KREMEN1	auto	ok	7	28.57	42.86	0.7034	no	0.2453
PLCG2	auto	ok	17	33.33	66.67	0.7133	no	0.2522
EXOC4	auto	ok	5	0	40	0.7162	no	0.5285
RAF1	auto	ok	23	25	50	0.7231	no	0.2143
ANKRD11	auto	ok	54	0	66.67	0.7555	no	0.4444
USP34	auto	ok	7	14.29	71.43	0.8018	no	0.4093
XPO1	auto	ok	31	8.333	41.67	0.8022	no	0.2217
ESR1	auto	ok	30	23.08	38.46	0.8411	no	0.2334
PIK3CB	auto	ok	20	0	60	0.8411	no	0.4483
CD79A	auto	ok	13	0	37.5	0.8411	no	0.4167
SDHA	auto	ok	12	40	60	0.8411	no	0.3269
NOTCH3	auto	ok	101	3.846	50	0.873	no	0.1881
FUBP1	auto	ok	29	18.18	54.55	0.873	no	0.2052
ARID1A	auto	ok	148	12.9	35.48	0.8816	no	0.2097
CIC	auto	ok	68	0	42.86	0.8816	no	0.3363
KDM5A	auto	ok	44	30	50	0.8816	no	0.2332
ERBB2	auto	ok	88	15.91	43.18	0.9017	no	0.2189
ATM	auto	ok	72	3.704	29.63	0.9017	no	0.2108

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
BRCA1	auto	ok	66	7.407	48.15	0.9017	no	0.1984
CACNA1A	auto	ok	7	14.29	42.86	0.9017	no	0.2188
NELL1	auto	ok	5	20	80	0.9017	no	0.3912
ARID5B	auto	ok	31	0	33.33	0.9223	no	0.4904
RB1	auto	ok	147	0	47.37	0.9236	no	0.3984
KMT2A	auto	ok	65	19.05	47.62	0.9236	no	0.1906
RAD21	auto	ok	38	44.44	55.56	0.9236	no	0.3412
PICALM	auto	ok	5	0	60	0.9236	no	0.4194
CSF3R	auto	ok	17	33.33	66.67	0.9261	no	0.2944
FOXA1	auto	ok	51	0	18.18	0.9568	no	0.5259
NF2	auto	ok	46	8.333	8.333	0.9568	no	0.2269
NOTCH4	auto	ok	44	0	68.75	0.9568	no	0.3691
SMARCA4	auto	ok	143	19.44	41.67	0.9955	no	0.2561
CCND1	auto	ok	15	0	57.14	0.9955	no	0.4286
NF1	auto	ok	247	10.64	44.68	1	no	0.2742
KMT2D	auto	ok	128	2.128	36.17	1	no	0.1893
WT1	auto	ok	119	89.42	4.808	1	no	0.6209
APC	auto	ok	110	0	45.45	1	no	0.446
EP300	auto	ok	98	4.545	31.82	1	no	0.2053
BRD4	auto	ok	86	3.125	34.38	1	no	0.1909
PTEN	auto	ok	86	22.22	22.22	1	no	0.2667
DOT1L	auto	ok	85	22.73	36.36	1	no	0.2478
PBRM1	auto	ok	85	5.556	50	1	no	0.2127
ZFHX3	auto	ok	85	0	14.29	1	no	0.5238
DNMT1	auto	ok	80	17.86	17.86	1	no	0.2299
BRCA2	auto	ok	79	8.333	58.33	1	no	0.214
ARID2	auto	ok	72	0	23.08	1	no	0.4156
CTNNB1	auto	ok	72	30	40	1	no	0.2134
KDM6A	auto	ok	71	25	75	1	no	0.08219
BAP1	auto	ok	67	16.67	33.33	1	no	0.2515
MGA	auto	ok	65	6.667	46.67	1	no	0.2372
ATRX	auto	ok	64	28.57	42.86	1	no	0.281
SMAD4	auto	ok	62	0	16.67	1	no	0.4259
SPEN	auto	ok	61	0	26.32	1	no	0.357
CDH1	auto	ok	58	11.11	66.67	1	no	0.2045
PTPRD	auto	ok	57	12.5	37.5	1	no	0.2176
RTEL1	auto	ok	57	20.83	37.5	1	no	0.2173

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
PIK3R1	auto	ok	55	33.33	33.33	1	no	0.153
MDC1	auto	ok	52	7.143	57.14	1	no	0.2447
GLI1	auto	ok	51	19.05	57.14	1	no	0.2434
STAT3	auto	ok	51	8.333	8.333	1	no	0.1969
FANCA	auto	ok	49	15	35	1	no	0.239
MTOR	auto	ok	49	10	30	1	no	0.2128
ATR	auto	ok	48	0	28.57	1	no	0.3882
BRIP1	auto	ok	46	0	21.05	1	no	0.3595
SETD2	auto	ok	46	0	47.06	1	no	0.3671
NSD1	auto	ok	45	8.333	33.33	1	no	0.2471
PTPRS	auto	ok	44	7.143	35.71	1	no	0.1952
PIK3C2G	auto	ok	43	30	30	1	no	0.2381
B2M	auto	ok	42	0	0	1	no	0.6666
ERBB4	auto	ok	42	40	40	1	no	0.2314
DNMT3B	auto	ok	40	0	14.29	1	no	0.3882
KEAP1	auto	ok	40	0	25	1	no	0.3785
NFE2L2	auto	ok	40	0	20	1	no	0.5001
EZH2	auto	ok	39	14.29	7.143	1	no	0.2783
FOXP1	auto	ok	39	20	80	1	no	0.2861
NBN	auto	ok	38	0	33.33	1	no	0.4434
AXL	auto	ok	37	30	20	1	no	0.2136
PREX2	auto	ok	37	0	50	1	no	0.4466
BCOR	auto	ok	36	0	57.14	1	no	0.3882
DROSHA	auto	ok	36	42.86	14.29	1	no	0.3074
IGF1R	auto	ok	36	8.333	25	1	no	0.2019
TBX3	auto	ok	36	0	50	1	no	0.4012
RICTOR	auto	ok	34	30	50	1	no	0.2787
KDM5C	auto	ok	33	30	60	1	no	0.2085
NCOA3	auto	ok	33	23.08	69.23	1	no	0.2059
PTPRT	auto	ok	33	33.33	50	1	no	0.221
RUNX1	auto	ok	33	16.67	33.33	1	no	0.2765
BCL2L11	auto	ok	32	21.43	21.43	1	no	0.3364
JAK3	auto	ok	32	0	33.33	1	no	0.392
PIK3R2	auto	ok	32	10	40	1	no	0.2632
RPS6KB2	auto	ok	32	33.33	0	1	no	0.2761
ERCC2	auto	ok	31	25	75	1	no	0.2651
INSR	auto	ok	31	16.67	50	1	no	0.2279

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
PDGFRA	auto	ok	31	55.56	55.56	1	no	0.224
TP53BP1	auto	ok	31	18.18	54.55	1	no	0.2741
ETV1	auto	ok	30	52.38	0	1	no	0.3978
GNAS	auto	ok	30	25	0	1	no	0.2779
AXIN1	auto	ok	29	50	66.67	1	no	0.4256
CALR	auto	ok	29	44.44	0	1	no	0.2806
CBL	auto	ok	29	25	50	1	no	0.3399
JAK1	auto	ok	29	12.5	0	1	no	0.2064
NTRK2	auto	ok	29	0	40	1	no	0.4405
AXIN2	auto	ok	28	0	75	1	no	0.4466
RARA	auto	ok	28	28.57	35.71	1	no	0.2158
TEK	auto	ok	28	0	55.56	1	no	0.3709
TET1	auto	ok	28	14.29	28.57	1	no	0.2135
MDM2	auto	ok	23	23.53	47.06	1	no	0.2584
MED12	auto	ok	23	0	33.33	1	no	0.4169
SMARCB1	auto	ok	23	0	60	1	no	0.5993
STAT5A	auto	ok	23	7.692	0	1	no	0.204
TCF7L2	auto	ok	23	0	50	1	no	0.5
EIF4A2	auto	ok	22	0	33.33	1	no	0.4012
PARP1	auto	ok	22	10	40	1	no	0.2584
PIK3CD	auto	ok	22	12.5	50	1	no	0.2195
PLK2	auto	ok	22	28.57	57.14	1	no	0.3264
PPP2R1A	auto	ok	22	50	33.33	1	no	0.2948
PRKCI	auto	ok	22	33.33	50	1	no	0.2899
RAD54L	auto	ok	22	0	60	1	no	0.4194
FLT3	auto	ok	21	0	20	1	no	0.4441
MAP2K2	auto	ok	21	28.57	14.29	1	no	0.2448
SMARCD1	auto	ok	21	12.5	50	1	no	0.2672
BLM	auto	ok	20	0	50	1	no	0.4135
FLCN	auto	ok	20	50	40	1	no	0.2682
MSH6	auto	ok	20	0	66.67	1	no	0.4012
PIK3R3	auto	ok	20	33.33	0	1	no	0.1419
SMYD3	auto	ok	20	25	25	1	no	0.1125
SRC	auto	ok	20	25	0	1	no	0.2485
CHEK2	auto	ok	19	16.67	16.67	1	no	0.2764
CSF1R	auto	ok	19	0	33.33	1	no	0.3709
GRIN2A	auto	ok	19	0	16.67	1	no	0.4012

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
LATS1	auto	ok	19	16.67	33.33	1	no	0.2232
PTCH1	auto	ok	19	0	20	1	no	0.4317
YAP1	auto	ok	19	20	20	1	no	0.2861
FGFR4	auto	ok	18	11.11	44.44	1	no	0.2053
PGR	auto	ok	18	14.29	57.14	1	no	0.2665
VTCN1	auto	ok	18	20	60	1	no	0.288
EGFL7	auto	ok	17	50	25	1	no	0.4086
ERCC5	auto	ok	17	0	37.5	1	no	0.5
LDLR	auto	ok	17	20	53.33	1	no	0.3759
SDC4	auto	ok	17	82.35	0	1	no	0.4731
TAP1	auto	ok	17	0	42.86	1	no	0.4286
TGFBR1	auto	ok	17	25	75	1	no	0.2714
CTCF	auto	ok	16	0	50	1	no	0.5
CUL1	auto	ok	16	35.71	0	1	no	0.3837
PDGFRB	auto	ok	16	40	40	1	no	0.2707
RECQL	auto	ok	16	0	0	1	no	0.4713
SMO	auto	ok	16	0	12.5	1	no	0.4032
SUZ12	auto	ok	16	0	20	1	no	0.4194
CCNE1	auto	ok	15	20	0	1	no	0.2861
MSH3	auto	ok	15	0	50	1	no	0.5
TNFRSF14	auto	ok	15	25	0	1	no	0.2625
CDK4	auto	ok	14	33.33	0	1	no	0.1183
E2F3	auto	ok	14	0	20	1	no	0.4194
FOXO1	auto	ok	14	50	0	1	no	0.3721
MPL	auto	ok	14	0	83.33	1	no	0.4665
NUF2	auto	ok	14	40	10	1	no	0.2888
NUP93	auto	ok	14	0	20	1	no	0.4194
SESN3	auto	ok	14	20	0	1	no	0.1214
SF3B1	auto	ok	14	20	40	1	no	0.2872
EZR	auto	ok	13	61.54	0	1	no	0.4151
MST1R	auto	ok	13	20	40	1	no	0.253
NPM1	auto	ok	13	50	50	1	no	0.3172
TRAPPC9	auto	ok	13	23.08	0	1	no	0.4889
YES1	auto	ok	13	16.67	16.67	1	no	0.2443
AKT3	auto	ok	12	33.33	33.33	1	no	0.267
CEBPA	auto	ok	12	0	0	1	no	0.4444
DUSP4	auto	ok	12	0	33.33	1	no	0.4135

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
FYN	auto	ok	12	66.67	66.67	1	no	0.1422
IDH1	auto	ok	12	25	0	1	no	0.2625
PAX5	auto	ok	12	33.33	66.67	1	no	0.186
PIK3C3	auto	ok	12	100	100	1	no	0.3772
PIK3CA	auto	ok	12	0	80	1	no	0.5222
ERCC4	auto	ok	11	20	40	1	no	0.2872
FGF19	auto	ok	11	0	0	1	no	0.4888
GSK3B	auto	ok	11	50	0	1	no	0.2625
MKRN1	auto	ok	11	63.64	0	1	no	0.3815
MLH1	auto	ok	11	50	50	1	no	0.2625
PHF7	auto	ok	11	50	0	1	no	0.2625
RAD51D	auto	ok	11	50	0	1	no	0.2625
RELA	auto	ok	11	33.33	100	1	no	0.426
SOS1	auto	ok	11	33.33	0	1	no	0.1465
TACC2	auto	ok	11	62.5	100	1	no	0.5485
BMPR1A	auto	ok	10	100	0	1	no	0.3772
CDK8	auto	ok	10	25	0	1	no	0.1472
CUL3	auto	ok	10	50	0	1	no	0.2599
EPAS1	auto	ok	10	0	20	1	no	0.4194
IKZF3	auto	ok	10	25	62.5	1	no	0.2719
MDM4	auto	ok	10	25	50	1	no	0.3137
RXRA	auto	ok	10	33.33	66.67	1	no	0.1419
TRIM24	auto	ok	10	88.89	0	1	no	0.3976
ARAF	auto	ok	9	50	50	1	no	0.2082
LRP1	auto	ok	9	0	20	1	no	0.4347
NTHL1	auto	ok	9	33.33	66.67	1	no	0.1961
PPARG	auto	ok	9	50	50	1	no	0.2072
RAD51	auto	ok	9	66.67	66.67	1	no	0.2773
SESN1	auto	ok	9	0	0	1	no	0.4194
STK40	auto	ok	9	33.33	0	1	no	0.1953
ETV4	auto	ok	8	33.33	16.67	1	no	0.3893
HNF1A	auto	ok	8	0	20	1	no	0.4441
IKZF1	auto	ok	8	100	0	1	no	0.3772
IRF4	auto	ok	8	0	20	1	no	0.4667
KRAS	auto	ok	8	33.33	0	1	no	0.1856
MAPK1	auto	ok	8	50	0	1	no	0.2599
MCL1	auto	ok	8	0	50	1	no	0.4538

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
MSI1	auto	ok	8	20	40	1	no	0.2192
MTAP	auto	ok	8	25	25	1	no	0.4351
MYD88	auto	ok	8	33.33	33.33	1	no	0.2253
PDE4D	auto	ok	8	16.67	66.67	1	no	0.3623
PTK2	auto	ok	8	25	50	1	no	0.276
SLC25A21	auto	ok	8	0	12.5	1	no	0.4466
STRN	auto	ok	8	25	0	1	no	0.5381
CDK6	auto	ok	7	50	0	1	no	0.2599
CRKL	auto	ok	7	0	50	1	no	0.6442
DCUN1D1	auto	ok	7	66.67	0	1	no	0.1606
DNM2	auto	ok	7	28.57	42.86	1	no	0.2286
GAB2	auto	ok	7	20	20	1	no	0.2861
IGF2	auto	ok	7	50	75	1	no	0.2614
ST7	auto	ok	7	14.29	0	1	no	0.437
TACC1	auto	ok	7	25	75	1	no	0.2625
CAMTA1	auto	ok	6	20	0	1	no	0.1241
CDKN2C	auto	ok	6	33.33	66.67	1	no	0.1594
CPM	auto	ok	6	40	0	1	no	0.1385
CTNNA3	auto	ok	6	60	0	1	no	0.2525
DTNB	auto	ok	6	16.67	0	1	no	0.3383
EIF3H	auto	ok	6	60	0	1	no	0.529
EPHX3	auto	ok	6	16.67	0	1	no	0.2625
NOS1AP	auto	ok	6	0	16.67	1	no	0.4434
SMARCA2	auto	ok	6	33.33	66.67	1	no	0.1424
SOX2	auto	ok	6	0	75	1	no	0.5009
TANC2	auto	ok	6	16.67	50	1	no	0.2263
TFG	auto	ok	6	100	100	1	no	0.3246
ZNRF3	auto	ok	6	33.33	66.67	1	no	0.4336
ATE1	auto	ok	5	33.33	0	1	no	0.3752
BCL2L1	auto	ok	5	50	50	1	no	0.2093
CNTNAP2	auto	ok	5	66.67	0	1	no	0.267
CREM	auto	ok	5	20	100	1	no	0.2602
CTTNBP2	auto	ok	5	0	20	1	no	0.5285
EIF4E	auto	ok	5	50	0	1	no	0.2599
FBXL7	auto	ok	5	50	75	1	no	0.4249
GPATCH8	auto	ok	5	25	50	1	no	0.3093
HCN1	auto	ok	5	100	0	1	no	0.3772

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
LRP1B	auto	ok	5	50	50	1	no	0.2008
MAD1L1	auto	ok	5	40	0	1	no	0.2382
PRKAG2	auto	ok	5	20	40	1	no	0.2774
RAC2	auto	ok	5	50	0	1	no	0.2599
SBNO2	auto	ok	5	0	75	1	no	0.4466
SDHAF2	auto	ok	5	100	0	1	no	0.4933
SFPQ	auto	ok	5	25	75	1	no	0.4008
SHANK2	auto	ok	5	0	25	1	no	0.4466
SMARCE1	auto	ok	5	100	0	1	no	0.4933
SMURF2	auto	ok	5	25	75	1	no	0.2569
SPTBN1	auto	ok	5	50	0	1	no	0.3932
THSD4	auto	ok	5	25	25	1	no	0.2439
TMPRSS2	auto	failed	1088					
CDKN2A	auto	ok	185	4.762	0			0.2857
KMT2B	unresolved	failed	68					
NSD3	auto	ok	58	14.29	0			0.1429
RECQL4	unresolved	failed	54					
PRKN	auto	ok	40	50	0			0.25
NSD2	auto	ok	37	33.33	0			0.3333
MAP3K14	unresolved	failed	36					
ASXL1	auto	ok	35	0	77.78			0.2222
TCF3	unresolved	failed	35					
RASA1	auto	ok	32	0	50			0.5
STAG2	auto	ok	31	0	0			0.5
AR	auto	ok	30	0	50			0.5
CDC73	auto	ok	28	0	50			0.5
IKBKE	auto	failed	27					
JAK2	auto	failed	27					
MAP2K4	auto	failed	27					
PPM1D	auto	failed	27					
SMAD3	auto	failed	27					
STAT5B	auto	failed	27					
TRAF7	auto	failed	27					
IRS2	auto	failed	26					
SOX9	auto	failed	26					
UPF1	auto	failed	26					
CARD11	auto	failed	25					

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
FLT1	auto	failed	25					
MALT1	auto	failed	25					
PAK1	auto	failed	25					
ASXL2	auto	failed	24					
CDKN1B	auto	failed	24					
DIS3	auto	failed	24					
TP63	auto	failed	24					
AKT1	auto	failed	23					
MAP3K13	auto	failed	23					
MSH2	auto	ok	23	0	0			0.2731
NFKBIA	auto	ok	23	0	100			1
MEN1	auto	ok	22	0	0			0.3016
MYC	auto	ok	22	0	0			0.3333
TET2	auto	ok	21	0	66.67			0.5516
ERF	unresolved	failed	20					
KIT	auto	ok	20	0	0			0.4266
CDKN2B-AS1	unresolved	failed	19					
EPHA3	auto	ok	19	0	0			0.4266
NEGR1	auto	ok	19	0	66.67			0.3333
PALB2	auto	ok	19	0	0			0.2
SEPTIN14	unresolved	failed	19					
BBC3	auto	ok	18	50	0			0.375
GATA3	auto	ok	18	0	0			0.5
AMER1	auto	ok	17	0	33.33			0.3333
COP1	auto	ok	17	50	0			0.5
DDR2	auto	ok	17	0	0			0.2016
EPHA5	auto	ok	17	0	33.33			0.5516
INPP4B	auto	ok	17	100	0			1
RAD50	auto	ok	17	0	33.33			0.3016
BCL6	auto	ok	16	0	66.67			0.3016
DICER1	auto	ok	16	0	0			0.3856
ERRFI1	auto	ok	16	0	0			0.6667
LATS2	auto	ok	16	0	0			0.8016
MITF	auto	ok	16	0	100			1
MSI2	auto	ok	16	0	50			0.4266
NCOA4	auto	failed	16					
RAD52	auto	ok	16	0	100			0.8016

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
TGFBR2	auto	ok	16	0	50			0.5
TRAP1	unresolved	failed	16					
CREB3L1	unresolved	failed	15					
INPP4A	auto	ok	15	14.29	0			0.2857
SRP19	auto	ok	15	0	0			1
GPS2	auto	ok	14	0	0			0.5
HGF	auto	ok	14	0	50			0.4606
LINC00114	unresolved	failed	14					
RIT1	auto	ok	14	0	0			0.3596
TAP2	auto	ok	14	0	0			0.5
TRAF2	unresolved	failed	14					
CYLD	auto	ok	13	0	0			1
DAXX	auto	ok	13	0	0			1
MAP2K1	auto	ok	13	0	50			0.25
MEF2B	unresolved	failed	13					
MRE11	auto	ok	13	0	20			0.2
PAK5	auto	ok	13	50	0			0.5
RAD51C	auto	ok	13	0	0			0.1766
SPOP	auto	ok	13	0	50			0.4266
BIRC3	auto	ok	12	0	66.67			0.3016
BTK	auto	ok	12	0	0			1
CBFB	auto	ok	12	0	0			1
EED	auto	ok	12	0	0			0.4266
IRS1	auto	ok	12	0	0			0.5
RAB35	auto	ok	12	0	0			0.3356
SMAD2	auto	ok	12	0	0			0.8016
VEGFA	auto	ok	12	0	0			0.4606
ALOX12B	auto	ok	11	0	0			0.4266
BABAM1	auto	ok	11	0	0			0.5
BCL2	auto	ok	11	0	0			1
EPHA7	auto	ok	11	0	50			0.4266
FANCC	auto	ok	11	0	0			0.6096
FBXW7	auto	ok	11	0	0			0.3016
IDH2	auto	ok	11	0	0			0.4266
SRSF2	auto	ok	11	0	100			1
TNFAIP3	auto	ok	11	0	0			0.3016
AURKA	auto	ok	10	0	33.33			0.5516

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
CCND2	auto	ok	10	0	60			0.3516
EPHB1	auto	ok	10	0	50			0.5
H3C2	unresolved	failed	10					
MAPK3	auto	ok	10	0	0			0.8016
PIK3CG	auto	ok	10	0	0			
PMS1	auto	ok	10	0	0			
PTPN11	auto	ok	10	0	0			0.3333
SH2B3	auto	ok	10	0	25			0.25
SOX17	auto	ok	10	0	0			0.5856
SYK	auto	ok	10	0	0			1
CHEK1	auto	ok	9	0	0			0.4266
GATA1	auto	ok	9	0	0			1
IFNGR1	auto	ok	9	0	33.33			0.3016
MAPKAP1	auto	ok	9	0	100			0.8016
PPP6C	auto	ok	9	0	0			1
PRDM14	auto	ok	9	0	0			
REL	auto	ok	9	0	0			
SHQ1	auto	ok	9	0	50			0.5
AKAP8	auto	ok	8	0	0			1
AURKB	auto	ok	8	0	33.33			0.3333
BARD1	auto	ok	8	0	0			
BCL2L14	auto	ok	8	50	0			0.75
CCND3	auto	ok	8	0	0			0.8356
CD79B	auto	ok	8	0	33.33			0.3016
ERCC3	auto	ok	8	0	33.33			0.3016
FGF3	auto	ok	8	0	0			0.3333
GNA11	auto	ok	8	0	0			1
ICOSLG	auto	ok	8	0	0			
LYN	auto	ok	8	0	0			0.2391
NBPF20	unresolved	failed	8					
PDCD1	auto	ok	8	0	0			1
RRAS	auto	ok	8	0	33.33			0.3333
XRCC2	auto	ok	8	0	0			1
ACP3	auto	ok	7	71.43	0			0.4286
AP1B1	auto	ok	7	0	0			1
ASIC2	auto	ok	7	16.67	0			0.3333
ATP1B2	auto	ok	7	0	0			1

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
DCBLD1	auto	ok	7	0	0			
EPCAM	auto	ok	7	0	0			0.5
H3C4	auto	ok	7	0	0			
INHA	auto	ok	7	0	0			1
JUN	auto	ok	7	0	0			
KLF4	auto	ok	7	0	0			
NKX3-1	auto	ok	7	0	0			1
PDPK1	auto	ok	7	0	100			1
RAB11FIP4	unresolved	failed	7					
RYBP	auto	ok	7	0	0			
SDHC	auto	ok	7	0	66.67			0.3016
SHOC2	auto	ok	7	0	100			1
TENT5C	auto	ok	7	0	0			
AGAP3	unresolved	failed	6					
CD274	auto	ok	6	0	0			1
CD276	auto	ok	6	0	0			
CDH3	unresolved	failed	6					
FGF4	auto	ok	6	0	50			0.5
GALNT11	auto	ok	6	0	33.33			1
GRB7	auto	failed	6					
H1-2	auto	failed	6					
H3-3B	auto	failed	6					
H3C6	unresolved	failed	6					
HOXB13	auto	failed	6					
IL7R	auto	ok	6	0	100			0.5
INHBA	auto	ok	6	0	50			0.4266

Showing the first 500 of 544 scanned genes.

Honorable mentions: highly ranked non-FDR-significant genes worth human review

Did NOT survive genome-wide FDR correction -- ranked by raw Fisher p-value among non-significant genes. Not a claim of significance.

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret.%
1	RET	0.0004197	0.09074	194	75.26	92.27
2	FGFR2	0.0004247	0.09074	136	79.41	84.56

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret.%
3	ALK	0.001917	0.2116	272	83.09	95.96
4	EGFR	0.01145	0.2368	55	50.91	74.55
5	BRAF	0.01337	0.2116	179	84.36	91.06
6	FGFR3	0.01948	0.2116	152	48.68	93.42
7	CDKN2B	0.02222	0.3606	12	16.67	16.67
8	NTRK1	0.04826	0.2116	78	41.03	82.05
9	INPPL1	0.04895	0.4733	14	28.57	50
10	CDK12	0.0677	0.2116	43	16.28	37.21
11	NAB2	0.1126	0.2116	72	34.72	68.06
12	KDM5A	0.119	0.8816	10	30	50
13	FH	0.1333	0.673	16	6.25	12.5
14	FGFR1	0.1807	0.5489	23	30.43	47.83
15	POLD1	0.1923	0.2116	16	12.5	37.5

Figures

manhattan